Alex Chen
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OBJECTIVE
Passionate AI/ML Engineer with strong mathematical foundations in machine learning algorithms, deep learning, an
EDUCATION
Bachelor of Science in Computer Science & Applied Mathematics
University of California, Berkeley | GPA: 3.88 / 4.0 | Graduated: May 2024
Relevant Coursework: Machine Learning (CS 189), Deep Learning for Computer Vision, Linear Algebra, Probability
TECHNICAL SKILLS
- Programming: Python, C++, SQL, Bash
- Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn, NumPy, Pandas, LangC
- Deep Learning & NLP: Large Language Models (LLMs), RAG pipelines, fine-tuning (LoRA, QLoRA), Attention Mec
- Machine Learning Foundations: Gradient Descent, Backpropagation, SVMs, Decision Trees, Random Forests, XG
- MLOps & Tools: Docker, Git, Weights & Biases, FastAPI, Ray, AWS EC2/S3
PROJECTS
1. RAG-Based Document Intelligence Engine (PyTorch, HuggingFace, FastAPI)
- Built an end-to-end question answering system over 10,000+ scientific research papers with semantic vector searc
- Reduced hallucination rate by 34% through reciprocal rank fusion (RRF) and dynamic context chunking with 15% c
- Deployed FastAPI microservice inside Docker container serving sub-150ms inference latency.
2. Neural Image Segmentation for Autonomous Driving (PyTorch, OpenCV)
- Implemented U-Net and DeepLabV3+ architectures for semantic segmentation on Cityscapes dataset.
- Optimized loss convergence using Focal Loss and Cosine Annealing learning rate schedule, reaching 78.4% mean
- Accelerated inference by 2.4x via TensorRT post-training INT8 quantization.
3. Distributed Gradient Boosting Classifier (Python, NumPy, Scikit-Learn)
- Engineered a lightweight gradient boosted tree engine from scratch implementing second-order Taylor expansion g
- Evaluated performance on imbalanced tabular datasets using stratified k-fold and PR-AUC metrics.
EXPERIENCE
Machine Learning Intern | NeuroTech Systems | June 2023 - August 2023
- Designed automated feature selection and hyperparameter optimization pipelines for tabular predictive models.
- Monitored model drift and prediction latency in production utilizing custom drift-detection statistical metrics.